

where β_1 and β_2 are the sample skewness and the sample kurtosis given by (3) and (4), respectively.

16) *Burr XII Distribution*: If g takes the form

$$g(\lambda) = kc\lambda^{c-1}(1+\lambda^c)^{-k-1}$$

for $\lambda > 0$, then one can write

$$f(y) = \frac{kc}{\sqrt{2\pi}} I$$

where

$$I = \int_0^\infty \frac{w^{kc-1/2} \exp(-wx)}{(1+w^c)^{k+1}} dw.$$

This integral can be reduced to the following explicit form if $c = 2$:

$$\begin{aligned} I = & x^{3/2} \Gamma\left(-\frac{3}{2}\right) {}_1F_2\left(k+1; \frac{7}{4}, \frac{5}{4}; -\frac{x^2}{4}\right) \\ & + \frac{1}{2} B\left(\frac{3}{4}, \frac{\frac{1}{2}+2k}{2}\right) {}_1F_2\left(\frac{\frac{1}{2}+2k}{2}; \frac{1}{2}, \frac{1}{4}; -\frac{x^2}{4}\right) \\ & - \frac{x}{2} B\left(\frac{1}{4}, \frac{2k+\frac{3}{2}}{2}\right) {}_1F_2\left(\frac{2k+\frac{3}{2}}{2}; \frac{3}{2}, \frac{3}{4}; -\frac{x^2}{4}\right). \end{aligned}$$

The moment estimators of k and c are the solutions of the equations

$$\begin{aligned} & \sqrt{\beta_1} \left\{ kB \left(k - \frac{2}{c}, 1 + \frac{2}{c}\right) - k^2 B^2 \left(k - \frac{1}{c}, 1 + \frac{1}{c}\right) \right\}^{3/2} \\ & = 2k^3 B^3 \left(k - \frac{1}{c}, 1 + \frac{1}{c}\right) - 3k^2 B \left(k - \frac{1}{c}, 1 + \frac{1}{c}\right) \\ & \quad \times B \left(k - \frac{2}{c}, 1 + \frac{2}{c}\right) + kB \left(k - \frac{3}{c}, 1 + \frac{3}{c}\right) \end{aligned}$$

and

$$\begin{aligned} & \beta_2 \left\{ kB \left(k - \frac{2}{c}, 1 + \frac{2}{c}\right) - k^2 B^2 \left(k - \frac{1}{c}, 1 + \frac{1}{c}\right) \right\}^2 \\ & = -3k^4 B^4 \left(k - \frac{1}{c}, 1 + \frac{1}{c}\right) + 6k^3 B^2 \left(k - \frac{1}{c}, 1 + \frac{1}{c}\right) \\ & \quad \times B \left(k - \frac{2}{c}, 1 + \frac{2}{c}\right) - 4k^2 B \left(k - \frac{1}{c}, 1 + \frac{1}{c}\right) \\ & \quad \times B \left(k - \frac{3}{c}, 1 + \frac{3}{c}\right) + kB \left(k - \frac{4}{c}, 1 + \frac{4}{c}\right), \end{aligned}$$

where β_1 and β_2 are the sample skewness and the sample kurtosis given by (3) and (4), respectively.

III. CONCLUSION

We have derived a collection of 16 flexible formulas and the associated estimation procedures for the inverse Gaussian transform distribution by choosing 16 flexible models for the distribution of the scale parameter given by $\lambda = \sigma^2$. It is expected these formulas could be useful with respect to the application areas mentioned by [1].

REFERENCES

- [1] T. I. Alecu, S. Voloshynovskiy, and T. Pun, "The Gaussian transform of distributions: definition, computation and application," *IEEE Trans. Signal Process.*, vol. 54, no. 8, pp. 2976–2985, Aug. 2006.
- [2] A. P. Prudnikov, Y. A. Brychkov, and O. I. Marichev, *Integrals and Series*. New York: Gordon and Breach, 1986, vol. 1.
- [3] I. S. Gradshteyn and I. M. Ryzhik, *Table of Integrals, Series, and Products*, 6th ed. San Diego, CA: Academic, 2000.

Spectral Analysis and Reconstruction for Periodic Nonuniformly Sampled Signals in Fractional Fourier Domain

Ran Tao, *Senior Member, IEEE*, Bing-Zhao Li, and Yue Wang

Abstract—Nonuniform sampling occurs in many applications due to imperfect timebase or random events. Periodic nonuniform sampling is a special type of nonuniform sampling. The properties and applications of periodic nonuniform sampling signals in traditional Fourier domain have been extensively studied, but so far none of the research papers covering the spectral analysis and reconstruction of periodic nonuniformly sampled signals in fractional Fourier domain have been published. This correspondence is to explore the spectral properties of periodic nonuniformly sampled signals associated with the fractional Fourier transform. First, based on the uniform fractional Fourier transform kernel, the general spectral representation of periodic nonuniformly sampled signals has been derived. Second, detailed analysis of periodic nonuniformly sampled chirp signals in the fractional Fourier domain have been performed. The results can be used to estimate the chirp rate and the sampling offsets. Finally, a more simple relationship between the discrete fractional spectrum of periodic nonuniformly sampled signals and the continuous spectrum has been obtained. Based on this relationship, the original continuous spectrum can be reconstructed from periodic nonuniformly sampled signals in fractional Fourier domain. In addition, the simulations are carried out to verify the correctness of the results.

Index Terms—Chirp signal, fractional Fourier transform (FrFT), nonstationary signal, periodic nonuniform sampling, spectral analysis.

NOMENCLATURE

$\mathbb{R}$	Field of real numbers.
FT	(Ordinary) Fourier transform.
FrFT	Fractional Fourier transform.
CFT	Continuous fractional Fourier transform.
DFrFT	Discrete fractional Fourier transform.
$\mathcal{F}^\alpha[x(t)](u)$	The CFT with angle α of a signal $x(t)$.
$\mathbf{F}^\alpha[x_s(t)](u)$	The DFrFT with angle α of a signal $x_s(t) = \sum_{n=-\infty}^{\infty} x(nT)\delta(t-nT)$.

I. INTRODUCTION

Nonstationary signal processing is well known in digital signal processing area and has evoked considerable interest in the literature. As a special kind of nonstationary signal, the chirp signal can be looked as a natural phenomena and occurs in many areas, for example, gravity waves, seismography, and bats navigation systems, but it is renowned from its use in signal processing areas like radar and sonar [1]–[5].

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Many useful tools have been proposed to process non-stationary signals, such as short time Fourier transform, wavelet transform and fractional Fourier transform [6]–[8]. Among all these signal processing tools, the fractional Fourier transform (FrFT) has proved to be a perfect tool in solving problems in quantum physics, optics, and non-stationary signal processing [7], [8], especially for chirp signal processing [2], [4], [7], [8]. The advantage of using the fractional Fourier domain is that traditional nonbandlimited signal in Fourier domain maybe bandlimited in the fractional Fourier domain for a certain value of angle α [9]; therefore, the traditional bandlimited-based processing method can be used in the fractional Fourier domain to obtain the desired results. The properties of the fractional Fourier transform have been investigated thoroughly in a number of research papers [7]–[16] and the uniform sampling theorem expansion for the fractional Fourier transform of bandlimited or time-limited signals have been derived in [9]–[13]. The discrete fractional Fourier transform (DFrFT), the discrete fractional Fourier series (DFrFS), and the discrete-time fractional Fourier transform (DTFrFT) have also been proposed by different researchers [14]–[16].

However, all of the above-mentioned computational methods and the sampling theorems about signals in the fractional Fourier domain [9]–[16] always assume that the digital signal at hand is obtained by sampling an analog signal at uniform time intervals. No results regarding the spectral analysis and reconstruction of nonuniformly sampled signals in the fractional Fourier domain have been reported in the literature so far. Simultaneously, in many practical applications, in the sampling of nonstationary signals, such as gravity waves, broadband chirp signals, and radar and sonar signals [2]–[5], the condition of the sampling theorem in the fractional Fourier domain proposed previously in [9]–[13] may not be satisfied by the current monolithic analog-to-digital (A/D) converter technologies. To resolve these issues, we can utilize interleaving/multiplexing techniques to extend the capabilities of the current monolithic A/D converter technology [17], [18]. Thus, nonuniform sampling might occur in these data acquisition systems due to imperfect sampling timebase of interleaving/multiplexing techniques, and the errors it introduced are usually dominating and cannot be ignored in perfect signal reconstruction. Also, in some applications, nonuniform sampling may be introduced intentionally to achieve certain desirable functions [17]. All of these actions are taken in the Fourier domain to solve the above problems, but they are only suitable for the stationary or bandlimited signals of the Fourier domain [17]–[21], and there are no results about spectral analysis and reconstruction for nonuniformly sampled signals in the fractional Fourier domain have been published as yet. It is therefore worthwhile and interesting to explore the spectral properties and reconstruction algorithm for nonuniformly sampled nonstationary signals in the fractional Fourier domain.

In this correspondence, we emphasize on a special case of nonuniform sampling pattern, which is known as recurrent nonuniform sampling, block sampling, or periodic nonuniform sampling [17]–[21]. In this form of sampling, the sampling points are divided into several groups of points. The groups have a recurrent period, which is denoted by MT , that is equal to M times the Nyquist period. Each period consists of M nonuniform sampling points. Denoting the points in one period by $t_n, n = 0, 1, \dots, M-1$, the complete set of sampling points

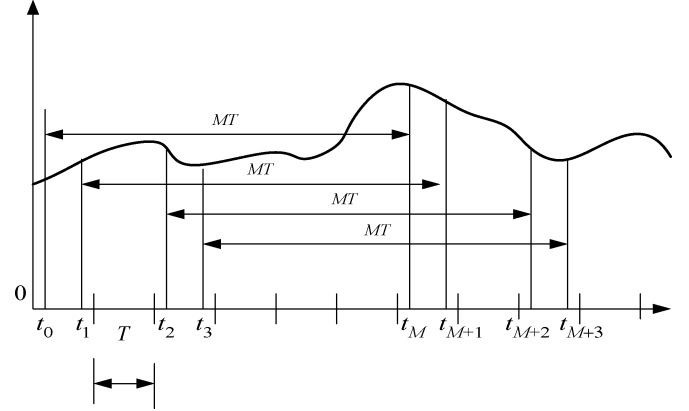


Fig. 1. Periodic nonuniform sampling model.

are $t_n + kMT, n = 0, 1, 2, \dots, M-1, k \in \mathbb{Z}$. Periodic nonuniform samples can be regarded as a combination of M sequences of uniform samples taken at one M th of the Nyquist rate. An example of this periodic nonuniform sampling distribution is depicted in Fig. 1 [17].

This correspondence is to investigate the spectral properties and reconstruction algorithm for periodic nonuniformly sampled signals in the fractional Fourier domain. The paper is organized as follows. In Section II, we review the definition of the fractional Fourier transform and obtain the fractional spectral representation of periodic nonuniformly sampled signals. Then, a detailed analysis of this representation about chirp signals is carried out. In Section III, a more simple relationship between the spectrum of periodic nonuniformly sampled signals and its continuous spectrum is deduced, the spectral reconstruction algorithm based on this relationship from periodic nonuniformly sampled signals in the fractional Fourier domain is proposed. Simulations are also carried out to verify the results in Section IV. Conclusions and future research directions along this way are proposed in Section V.

II. DFrFT OF NONUNIFORMLY SAMPLED SIGNALS

A. Preliminary

The CFrFT of a signal $f(t)$ with angle α is defined as

$$\mathcal{F}^\alpha[f(t)](u) = \mathcal{F}^\alpha(u) = \int_{-\infty}^{\infty} K_\alpha(u, t) f(t) dt \quad (1)$$

where (see the equation at the bottom of the page).

A signal $f(t)$ is called Ω_α bandlimited signal in the fractional Fourier transform sense, which means

$$\mathcal{F}^\alpha(u) = 0 \text{ for } |u| > \Omega_\alpha \quad (2)$$

where Ω_α is called the bandwidth of signal $f(t)$ in fractional Fourier domains.

$$K_\alpha(u, t) = \begin{cases} A_\alpha \exp\left(j\frac{t^2}{2} \cot \alpha - jut \csc \alpha + j\frac{u^2}{2} \cot \alpha\right), & \alpha \neq k\pi \\ \delta(t - u), & \alpha = 2k\pi \\ \delta(t + u), & \alpha + \pi = 2k\pi \end{cases}, \quad A_\alpha = \sqrt{\frac{1 - j \cot \alpha}{2\pi}}.$$

B. DFrFT of Periodic Nonuniformly Sampled Signal

In the above-mentioned periodic nonuniform sampling model, the m th sampling sequence can be seen as uniformly sampling the signal with sampling period MT , as follows:

$$s_m = [f(t_m), f(t_{M+m}), f(t_{2M+m}), \dots] \quad (3)$$

and the total sampling sequence can be represented as

$$s = \{f(t_{km}) | t_{km} = kMT + t_m, m = 0, 1, 2, \dots, M-1, k \in \mathbb{Z}\}. \quad (4)$$

Let $\bar{s}_m = [f(t_m), 0, \dots, (M-1)\text{zeros}, f(t_{M+m}), 0, 0, \dots]$, $m = 0, 1, \dots, M-1$ and shift $\bar{s}_m mT$ positions to the right, for $m = 0, 1, \dots, M-1$ to obtain

$$\bar{s}_m z^{-m} = [(m \text{ zeros})f(t_m), 0, \dots, (M-1)\text{zeros}, f(t_{M+m}), 0, 0, \dots] \quad (5)$$

where z^{-1} is the unit delay operator. Finally, summing up all the subsequences to obtain the original sequence $s = \sum_{m=0}^{M-1} \bar{s}_m z^{-m}$, therefore the discrete fractional spectrum of the periodic nonuniformly sampled signal sequence s can be derived by the summation of the FrFT of M subsequence $\bar{s}_m z^{-m}$ ($m = 0$ to $M-1$). The result can be represented as the following theorem.

Theorem 1: Let $f(t)$ be an analog signal with its CFrFT $\mathcal{F}^\alpha(u)$ bandlimited to $(-\Omega_\alpha, \Omega_\alpha)$ in the α th fractional Fourier domain; $f(t)$ is sampled nonuniformly with recurrent period MT . Then, the fractional spectrum of these periodic nonuniformly sampled points can be represented as

$$\begin{aligned} \mathbf{F}^\alpha[s](u) &= \frac{1}{MT} \\ &\times \sum_{n=-\infty}^{\infty} \sum_{m=0}^{M-1} \mathcal{F}^\alpha \left[u - n \frac{2\pi \sin \alpha}{MT} - r_m T \cos \alpha \right] \\ &\cdot e^{j \frac{1}{2} T^2 r_m^2 \sin \alpha \cos \alpha} e^{j \frac{2\pi n}{MT} \cos \alpha (u - \frac{\pi n \sin \alpha}{MT})} \\ &\times e^{-j r_m T \sin \alpha (u - \frac{2\pi n \sin \alpha}{MT})} e^{-j \frac{2\pi n m}{M}} \end{aligned} \quad (6)$$

where T is the nominal uniform sampling period, and $r_m = (mT - t_m)/T$, $m = 0, 1, \dots, M-1$ are sampling time offsets.

Proof: The results can be derived by utilizing the properties of FrFT and complex computations. ■

It is easy to show that the results of Theorem 1 reduce to the well-known results [17] in the Fourier domain when $\alpha = \pi/2$. Furthermore, our results can also be used for spectral analysis of periodic nonuniformly sampled chirp signals. Therefore, the properties and potential applications of this result for chirp signal processing are investigated in the following.

C. Fractional Spectrum of Chirp Signals

The CFrFT of a chirp signal $f(t) = C e^{j(w_0 t + (1/2)m_0 t^2)}$ with angle α ($\cot \alpha + m_0 = 0$, C is a constant) can be derived as follows:

$$\mathcal{F}^\alpha(u) = 2\pi C A_\alpha e^{j \frac{u^2}{2} \cot \alpha} \delta(w_0 - u \csc \alpha) \quad (7)$$

Therefore, the second theorem of this paper can be derived as:

Theorem 2: Let $f(t) = C e^{j(w_0 t + (1/2)m_0 t^2)}$, then the α th ($\cot \alpha + m_0 = 0$, C is a constant) fractional spectrum $\mathcal{F}^\alpha(u)$ from the periodic nonuniformly sampled signal points with recurrent period MT can be represented by the following equations:

$$\mathbf{F}^\alpha(u) = \frac{2\pi C A_\alpha}{T} e^{j \frac{1}{2} \cot \alpha u^2} \sum_{k=-\infty}^{\infty} A(k) \delta \left[w_0 - u \csc \alpha + k \frac{2\pi}{MT} \right] \quad (8)$$

where

$$A(k) = \frac{1}{M} \sum_{m=0}^{M-1} \left[e^{-j r_m T (w_0 + \frac{1}{2} \cot \alpha T r_m) - j M T r_m T \cot \alpha} \right] e^{-j m k \frac{2\pi}{M}}. \quad (9)$$

Proof: This relationship can be deduced by substituting (7) in (6), and applying the properties of discrete Fourier transform. ■

1) **Remarks:**

- Equation (8) indicates that the fractional spectrum of a chirp signal from periodic nonuniformly sampled points contains line spectra components in the α th fractional Fourier domain, with the magnitude of the k th component being $A(k)$.
- The spectra of periodic nonuniformly sampled signals have the same characteristic as the spectral of uniformly sampled signals. The only difference is that for periodic nonuniformly sampling signals, it will produce some additional pseudo spectra in the same α th fractional Fourier domain besides the main spectral component.
- From (9), the magnitude $A(k)$ can be seen as the discrete Fourier transform (DFT) of $e^{-j r_m T (f_0 + (1/2) \cot \alpha T r_m) - j M T r_m T \cot \alpha}$, it is periodic with k having period M . If the chirp signal satisfies the chirp periodicity of the order of α [13], then the fractional spectrum of the nonuniformly sampled signal is periodic with u having the period equal to $2\pi \sin \alpha / T$ in the chirp-periodicity sense of the fractional Fourier domain. Furthermore, one period of the spectrum comprises M line fractional spectra uniformly spaced with the α th fractional domain, with the neighboring spectra lines separated by the quantity of $\sin \alpha / MT$ with the main spectral component located at $f_0 \sin \alpha$, having the magnitude $|A(0)|$.

2) **Multiple Chirp Signals:** A multicomponent chirp signal can be modeled as $x(t) = \sum_{i=1}^N e^{j(f_i t + (1/2)m_i t^2)}$. Different m_i ($i = 1, 2, \dots, N$) will produce distinct line spectra in different fractional Fourier domains [7], [8]. Following this result and the properties of FrFT, the following corollary can be deduced directly.

Corollary 1: For a multiple chirp signal $f(t) = \sum_{l=1}^L \sum_{i=1}^N e^{j(w_{l,i} t + (1/2)m_i t^2)}$, the fractional spectrum of periodic nonuniformly sampled signal points with recurrent period MT in fractional Fourier domain can be represented as

$$\begin{aligned} \mathbf{F}^\alpha(u) &= \sum_{l=1}^L \sum_{i=1}^N \sum_{k=-\infty}^{\infty} \frac{2\pi C A_{\alpha i}}{T} e^{j \frac{1}{2} \cot \alpha_i u^2} A_{l,i}(k) \delta \\ &\times \left[w_{l,i} - u \csc \alpha_i + k \frac{2\pi}{MT} \right] \end{aligned} \quad (10)$$

where T is the nominal uniform sampling period, and

$$\begin{aligned} A_{l,i}(k) &= \frac{1}{M} \sum_{m=0}^{M-1} \left[e^{-j r_m T (f_{l,i} + \frac{1}{2} \cot \alpha_i T r_m) - j M T r_m T \cot \alpha_i} \right] \\ &\times e^{-j m k \frac{2\pi}{M}} \quad l = 1, 2, \dots, L, \quad i = 1, 2, \dots, N. \end{aligned} \quad (11)$$

3) **Potential Applications:**

- From the remarks a) and b) of Theorem 2, it is shown that the spectrum of periodic nonuniformly sampled signals contains line spectra components in the fractional Fourier domain just like the uniformly sampled signal performance at the same angle α besides the pseudo additional spectrum; therefore, the well known chirp rate estimation methods [4], [7] based on uniformly sampled signals can also be used to estimate the chirp rate from periodic nonuniformly sampled chirp signals.
- The coefficient of each spectra $A(k)$ can be seen as the discrete Fourier transform of $e^{-j r_m T (f_0 + (1/2) \cot \alpha T r_m) - j M T r_m T \cot \alpha}$; therefore, if the values of $A(k)$ in one period are known, the

sampling offsets r_m can be estimated by applying the inverse DFT of $A(k)$. Although this method works well in estimating the sampling jitters, in real situations the Fourier spectrum (when $\alpha = \pi/2$ in (8)) is used because the jitters are equipment errors. Whether $\alpha = \pi/2$ or not, this provides us with an alternative method to estimate the sampling offsets r_m besides the traditional method.

The other fundamental problem of signal processing is about the spectrum reconstruction from the uniformly or nonuniformly sampled signals. But from (6), we know that $\mathbf{F}^\alpha[s](u)$ is too complex for us to reconstruct the original spectrum $\mathcal{F}^\alpha(u)$; therefore, a simple relationship between the spectrum of nonuniformly sampled signal and its continuous spectrum based on the nonuniform fractional Fourier transform kernel $K_\alpha(u, t_n)$, $n \in Z$ has been derived in the following section.

III. SPECTRUM RECONSTRUCTION

A. Nonuniform Fractional Fourier Transform

For periodic nonuniform sampling time t_n , another fractional spectrum $G^\alpha(u)$ by the definition of DFrFT is derived as

$$G^\alpha(u) = \sum_{n=-\infty}^{+\infty} f(t_n) K_\alpha(u, t_n) \quad (12)$$

where $t_n = kMT + mT - r_mT$, $n = kM + m$, with k ranging from $-\infty$ to $+\infty$ and m from 0 to $M - 1$. Simultaneously, $f(t_n)$ can be represented as

$$f(t_n) = \int_{-\infty}^{+\infty} \mathcal{F}^\alpha(v) K_\alpha^*(v, t_n) dv. \quad (13)$$

Substituting (13) and t_n to (12), and obtaining

$$\begin{aligned} G^\alpha(u) &= \frac{1}{T \sin \alpha} e^{j \frac{1}{2} \cot \alpha u^2} \\ &\times \sum_{k=-\infty}^{+\infty} B(k) e^{-j \frac{1}{2} \cot \alpha \left[\left(u - k \frac{2\pi \sin \alpha}{MT} \right)^2 \right]} \\ &\times \mathcal{F}^\alpha \left(u - k \frac{2\pi \sin \alpha}{MT} \right) \end{aligned} \quad (14)$$

where

$$B(k) = \frac{1}{M} \sum_{m=0}^{M-1} e^{-jk \frac{2\pi}{M} (m+r_m)}. \quad (15)$$

Equation (14) introduces a simple relationship between the digital spectrum, $G^\alpha(u)$, and the original spectrum $\mathcal{F}^\alpha(u)$. It shows in (15) that $B(k)$ is not a function of u ; this fact can be used to deduce the reconstruct algorithm of original spectrum from periodic nonuniformly sampled signals.

B. Fractional Spectral Reconstruction

Assume the FrFT of a signal $f(t)$ bandlimited to $(-\pi \sin \alpha/T, \pi \sin \alpha/T)$ in α th fractional Fourier domain. Therefore, it will not be a bandlimited signal in the traditional Fourier domain if $\alpha \neq \pi/2 \pm n\pi$ [13], and thus the spectrum reconstruction algorithm proposed in [22] will not be suitable to this type signals. To overcome this shortcoming, the DFrFT of periodic nonuniformly sampled bandlimited signals based on (13) is performed in the fractional Fourier domain. Then, the following matrix relationship between digital spectrum $G_\alpha(u)$ and the original spectrum $\mathcal{F}^\alpha(u)$ is deduced:

$$\mathbf{G}^\alpha(u_0) = \mathbf{B} \bar{\mathcal{F}}^\alpha(u_0) \quad (16)$$

where (see the equation at the bottom of the page). From (16), the M uniformly sampled points of original signal spectrum $\mathcal{F}^\alpha(u_0)$ can be obtained by the following equation:

$$\bar{\mathcal{F}}^\alpha(u_0) = \mathbf{B}^{-1} \mathbf{G}^\alpha(u_0). \quad (17)$$

Finally, by choosing different values of u_0 carefully, we can derive enough uniformly sampled points of original signal spectrum $\mathcal{F}^\alpha(u)$.

IV. SIMULATION RESULTS

A. One Chirp Signal

The DFrFT of the uniformly sampled chirp signal $x(t) = e^{j(-6t^2+5t)}$ in the fractional domain based on the computational method proposed in [14] is plotted in Fig. 2; the nonuniformly sampled signal with recurrent period $M = 3$ in the same fractional domain is plotted in Fig. 3. It illustrates that when performing uniform

$$\begin{aligned} 0 < u_0 < \frac{2\pi \sin \alpha}{MT}, \\ \mathbf{B} &= \begin{pmatrix} B(M/2) & B(M/2-1) & \cdots & B(-M/2+1) \\ B(M/2+1) & B(M/2) & \cdots & B(-M/2+2) \\ B(M/2+2) & B(M/2+1) & \cdots & B(-M/2+3) \\ \vdots & \vdots & \vdots & \vdots \\ B(M/2+M-1) & B(M/2+M-2) & \cdots & B(M/2) \end{pmatrix} \\ \mathbf{G}_\alpha(u_0) &= \begin{pmatrix} G_\alpha(u_0) T \sin \alpha e^{-j \frac{1}{2} \cot(\alpha) u_0^2} \\ G_\alpha(u_0 + \sin \alpha 2\pi/MT) T \sin \alpha e^{-j \frac{1}{2} \cot(\alpha) (u_0 + 2\pi \sin \alpha/MT)^2} \\ \vdots \\ G_\alpha(u_0 + (M-1)2\pi \sin \alpha/MT) T \sin \alpha e^{-j \frac{1}{2} \cot(\alpha) (u_0 + (M-1)2\pi \sin \alpha/MT)^2} \end{pmatrix}, \\ \bar{\mathcal{F}}^\alpha(u_0) &= \begin{pmatrix} \mathcal{F}^\alpha\left(u_0 - \frac{\pi \sin \alpha}{T}\right) \\ \mathcal{F}^\alpha\left(u_0 - \frac{\pi \sin \alpha}{T} + \frac{2\pi \sin \alpha}{MT}\right) \\ \vdots \\ \mathcal{F}^\alpha\left(u_0 - \frac{\pi \sin \alpha}{T} + \frac{(M-1)2\pi \sin \alpha}{MT}\right) \end{pmatrix} \end{aligned}$$

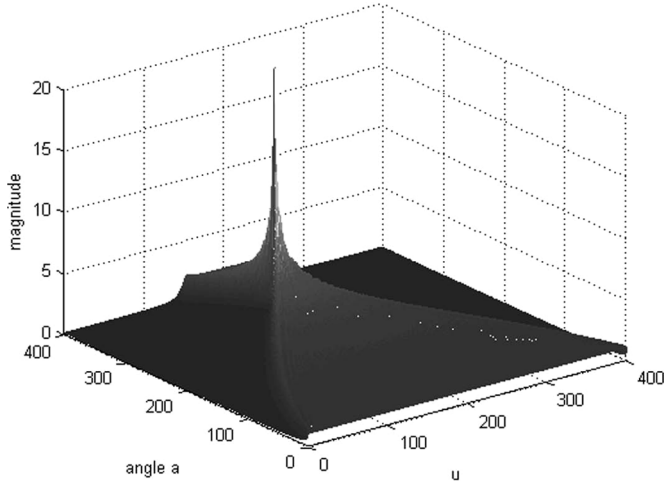
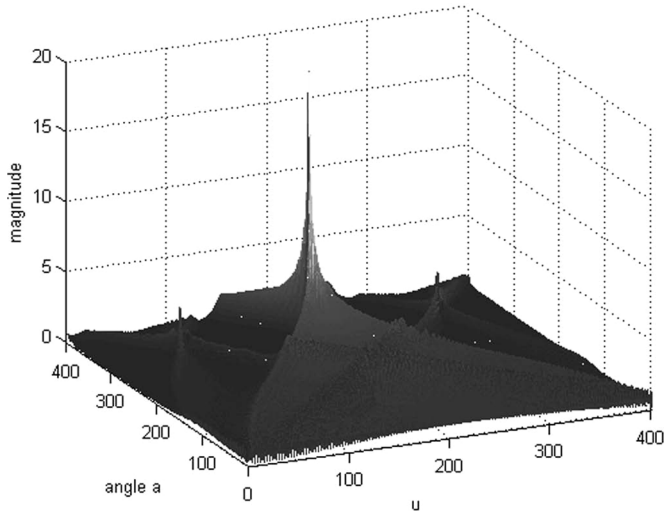


Fig. 2. DFrFT of uniformly sampled chirp signal.

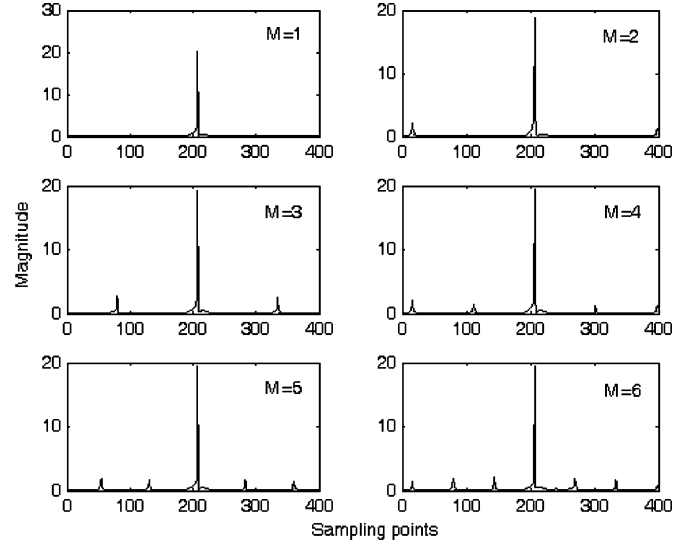
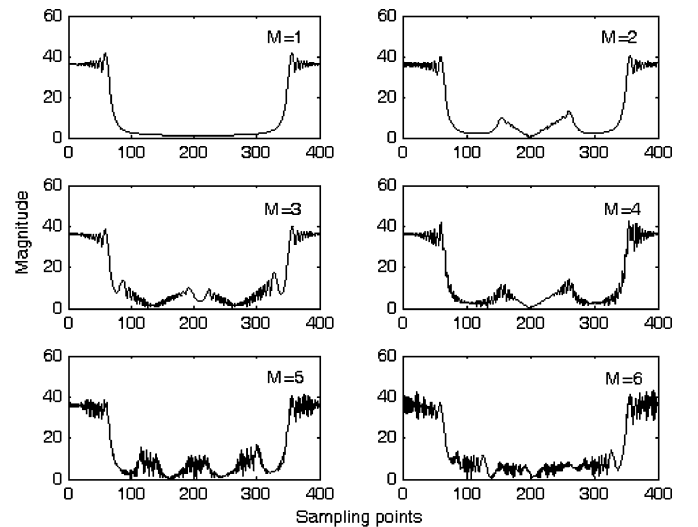
Fig. 3. DFrFT of nonuniformly sampled chirp signal with $M = 3$.

sampling, it produces only one line spectra in certain α th fractional Fourier domain, but when sampled nonuniformly with recurrent period M , there is an introduction of another $M - 1$ spectra lines besides the main spectrum component at the same angle α in the fractional Fourier domain.

In order to show the advantage of using the fractional Fourier domain for the spectrum analysis of chirp signal and the above phenomenon more explicitly, we exploit it in detail for angle α and compare the simulation results with the Fourier transform performance. The magnitudes of fractional spectrum of nonuniformly sampled signal at different recurrent period $M = 1, 2, 3, 4, 5, 6$, have been plotted, respectively, at angle $\alpha = 0.8112$ in Fig. 4. Fig. 5 plots the Fourier domain results. It is clear from Fig. 4 that one period of the spectrum comprises M line spectra uniformly spaced in the α th fractional Fourier domain with neighboring spectral lines separated by a certain number of points. In contrast to this, it will not obtain clear and simple results by the Fourier transform as reflected in Fig. 5.

B. Multiple Chirp Signals

In order to show that our method also works well with multiple chirp signals, a multiple chirp signal shown in (18) is considered. Figs. 6

Fig. 4. DFrFT of nonuniformly sampled signal at $\alpha = 0.8112$ for different period M .Fig. 5. DFT of nonuniformly sampled signal for different period M .

and 7 plot the spectrum of uniformly and periodic nonuniformly (with $M = 3$) sampled chirp signal, respectively.

$$x(t) = e^{j(-6t^2+5t)} + e^{j(-16t^2+2t)}. \quad (18)$$

It is interesting to see that the spectrum of uniformly sampled multiple chirp signal comprises N line spectra (in simulation example $N = 2$) in the fractional Fourier domain. But, for the nonuniformly sampled chirp signal, besides the main line spectra, it also introduces some pseudo line spectra in the fractional Fourier domain. And this fits the results well.

C. Spectrum Reconstruction

In simulation, the recurrent period is chosen as $M = 8$ and 512 uniform points of original spectrum in the α th fractional Fourier domain are obtained. The original fractional spectrum is plotted in Fig. 8; Fig. 9 plots the reconstructed fractional spectrum by the algorithm proposed

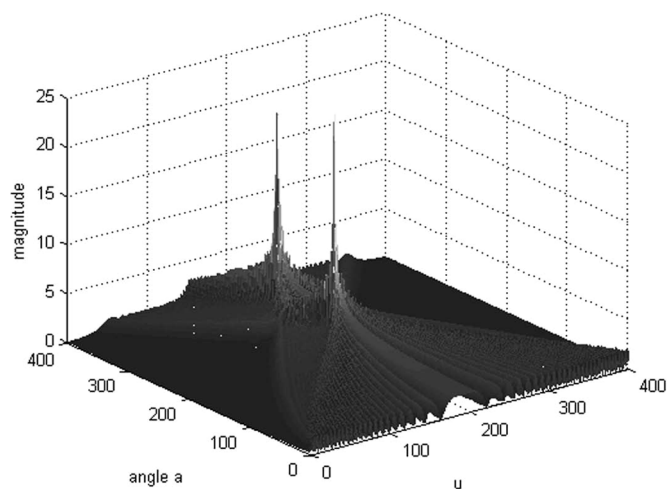
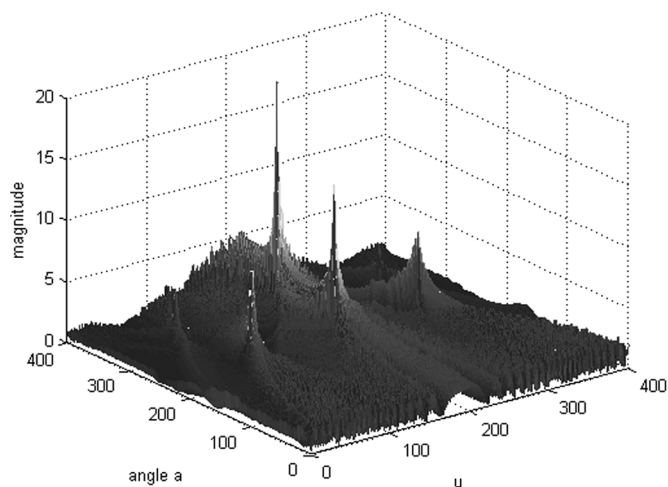
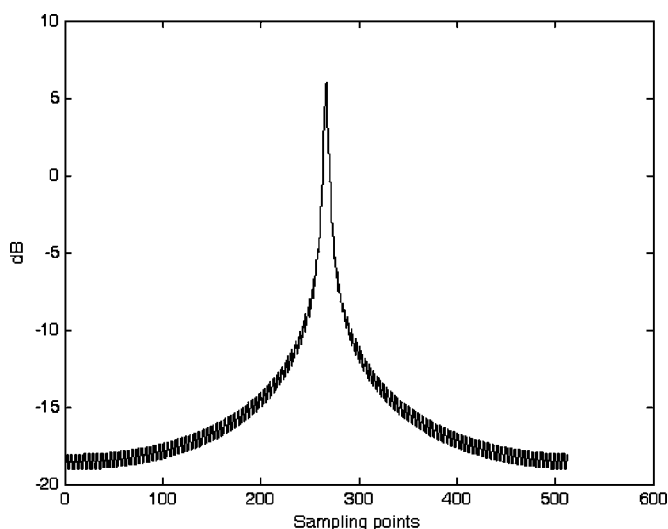
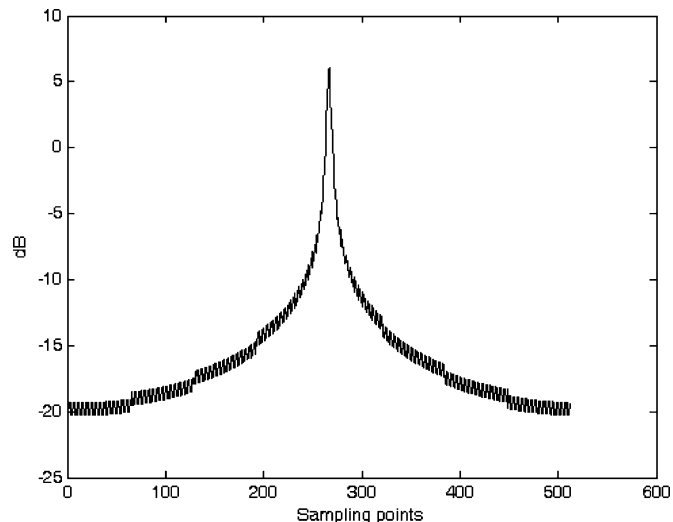


Fig. 6. DFrFT of uniformly sampled two chirp signals.

Fig. 7. DFrFT of nonuniformly sampled two chirp signals at $M = 3$.Fig. 8. Spectrum of original signal at $\alpha = 0.0875$.

from periodic nonuniformly sampled signals. The simulation results show that the algorithm works well.

Fig. 9. Reconstructed spectrum at $\alpha = 0.0875$ with $M = 8$.

V. CONCLUSION

This paper has investigated the spectral analysis and reconstruction of periodic nonuniformly sampled signals in the fractional Fourier domain. First, a general fractional spectral representation of periodic nonuniformly sampled signals in the fractional Fourier domain has been obtained, and a detailed analysis for chirp signals has been carried out. By using the achieved results, the chirp rate and the sampling offsets can be estimated. In order to reconstruct the original spectrum from periodic nonuniformly sampled signals, a more simple relationship between the discrete and continuous fractional spectrum have been deduced, and the spectral reconstruction algorithm from periodic nonuniformly sampled signals has also been proposed. In addition, simulation results for spectral analysis and reconstruction for nonuniformly sampled signals were proposed to verify the results. The future research and work will be in the direction of the theories for high-dimensional signals and applications of these results in high-speed A/D conversion analysis, array signal processing, and DOA estimation for nonstationary signals in fractional Fourier domain.

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REFERENCES

- [1] S.-C. Pei and J.-J. Ding, "Relations between fractional operations and time-frequency distributions, and their applications," *IEEE Trans. Signal Process.*, vol. 49, no. 8, pp. 1638–1655, Aug. 2001.
- [2] R. Tao, B. Deng, and Y. Wang, "Research progress of the fractional Fourier transform in signal processing," *Sci. in China (Ser. F, Inf. Sci.)*, vol. 49, no. 1, pp. 1–25, Jun. 2006.
- [3] O. Besson, M. Ghogho, and A. Swami, "Parameter estimation for random amplitude chirp signals," *IEEE Trans. Signal Process.*, vol. 47, no. 12, pp. 3208–3219, Dec. 1999.
- [4] L. Qi, R. Tao, S. Zhou, and Y. Wang, "Detection and parameter estimation of multicomponent LFM signal based on the fractional Fourier transform," *Sci. in China (Ser. F, Inf. Sci.)*, vol. 47, no. 2, pp. 184–198, Apr. 2004.
- [5] M. I. Skolnik, *Radar Handbook*. New York: McGraw-Hill, Jan. 1990.

- [6] I. Daubechies, *Ten Lectures on Wavelets*. Philadelphia, PA: SIAM, 1992.
- [7] H. M. Ozaktas, Z. Zalevsky, and M. A. Kutay, *The Fractional Fourier Transform With Applications in Optics and Signal Processing*. New York: Wiley, 2001.
- [8] R. Tao, L. Qi, and Y. Wang, *Theory and Applications of the Fractional Fourier Transform*. Beijing, China: Tsinghua Univ. Press, 2004.
- [9] X. Xia, "On bandlimited signals with fractional Fourier transform," *IEEE Signal Process. Lett.*, vol. 3, no. 3, pp. 72–74, Mar. 1996.
- [10] A. I. Zayed, "On the relationship between the Fourier and fractional Fourier transform," *IEEE Signal Process. Lett.*, vol. 3, no. 12, pp. 310–311, Dec. 1996.
- [11] W.-Q. Zhang and R. Tao, "Sampling theorems for bandpass signals with fractional Fourier transform," *Acta Electron. Sin.*, vol. 33, no. 7, pp. 1196–1199, Jul. 2005.
- [12] C. Candan and H. M. Ozaktas, "Sampling and series expansion theorems for fractional Fourier and other transforms," *Signal Process.*, vol. 83, pp. 2455–2457, 2003.
- [13] T. Erseghe, P. Kraniakauskas, and G. Cariolaro, "Unified fractional Fourier transform and sampling theorem," *IEEE Trans. Signal Process.*, vol. 47, no. 12, pp. 3419–3423, Dec. 1999.
- [14] H. M. Ozaktas, O. Arikan, M. A. Kutay, and G. Bozdagi, "Digital computation of the fractional Fourier transform," *IEEE Trans. Signal Process.*, vol. 44, no. 9, pp. 2141–2149, Sep. 1996.
- [15] S.-C. Pei, M.-H. Yeh, and C.-C. Tseng, "Discrete fractional Fourier transform based on orthogonal projections," *IEEE Trans. Signal Process.*, vol. 47, no. 5, pp. 1335–1348, May 1999.
- [16] C. Candan, M. A. Kutay, and H. M. Ozaktas, "The discrete fractional Fourier transform," *IEEE Trans. Signal Process.*, vol. 48, no. 5, pp. 1329–1337, May 2000.
- [17] Y.-C. Jenq, "Digital spectra of nonuniformly sampled signals: Fundamentals and high-speed waveform digitizers," *IEEE Trans. Instrum. Meas.*, vol. 37, no. 2, pp. 245–251, Jun. 1988.
- [18] J. L. Yen, "On nonuniform sampling of bandwidth-limited signals," *IEEE Trans. Circuit Theory*, vol. CT-3, no. 12, pp. 251–257, Dec. 1956.
- [19] A. Papoulis, "Generalized sampling expansion," *IEEE Trans. Circuit Syst.*, vol. cas-24, no. 11, Nov. 1977.
- [20] Y. C. Eldar and A. V. Oppenheim, "Filterbank reconstruction of bandlimited signals from nonuniform and generalized samples," *IEEE Trans. Signal Process.*, vol. 48, no. 10, pp. 2864–2875, Oct. 2000.
- [21] A. Feuer and G. C. Goodwin, "Reconstruction of multidimensional bandlimited signals from nonuniform and generalized samples," *IEEE Trans. Signal Process.*, vol. 53, no. 11, pp. 4273–4282, Nov. 2005.
- [22] Y.-C. Jenq, "Perfect reconstruction of digital spectrum from nonuniformly sampled signals," *IEEE Trans. Instrum. Meas.*, vol. 46, no. 3, pp. 649–652, Jun. 1997.

Error Analysis and Bounds in Time Delay Estimation

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Abstract—We have analyzed the error propagation in the classical delay estimation based on the identification of the maximum of the cross correlation between the input signals. The resulting error propagation model has the form of a linear combination of the errors additional to the signal samples. These errors are weighted by the derivative of the input signals at the sampling instants. Based on the error propagation model we have further found an upper bound to the square of the estimation error. The use of the model is demonstrated by two simple examples.

Index Terms—Delay estimation, error analysis, error propagation.

I. INTRODUCTION

The objective of the time delay estimation is to determine the delay D between a signal $s_1(t)$ and its scaled delayed version $s_2(t) = a s_1(t - D)$ where a is the scaling factor. We consider the use of the classical delay estimator based on the identification of the maximum of the cross correlation between the input signals. This can be found using the maximum likelihood approach and is widely discussed for the case of random noise [1]–[11].

We will focus on looking for a deterministic model of the error propagation in this estimator as a tool for particular delay estimation error analysis. For discrete time processing we assume continuous real signals which are effectively band-limited to $f_m < f_s/2$ where f_s is the sampling frequency. Note that the same results we would get also for undersampled narrowband signals.

For the purpose of the analysis we will use the following model of the estimation process. At the input of the estimator we have samples

$$x_1(kT) = s_1(kT) + \varepsilon_1(kT), \quad (1)$$

$$x_2(kT) = s_2(kT) + \varepsilon_2(kT), \quad (2)$$

where k is the discrete time index, $T = 1/f_s$ is the sampling period, and $\varepsilon_1(kT)$, $\varepsilon_2(kT)$ are additive errors. First, the continuous signals $x_1(t)$, $x_2(t)$ are reconstructed from all nonzero samples using an ideal interpolator. After that the estimation of the delay is determined as

$$\hat{D} = \arg \max R(\tau) \quad (3)$$

where $R(\tau)$ is the cross correlation formally defined as

$$R(\tau) = \int_{-\infty}^{\infty} x_1^*(t) x_2(t + \tau) dt \quad (4)$$

where the asterisk denotes complex conjugation.

II. ERROR PROPAGATION

We will find the error propagation from additive errors $\varepsilon_1(kT)$, $\varepsilon_2(kT)$ to the time delay estimation error

$$\Delta = \hat{D} - D. \quad (5)$$

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